



Edition 2026.1

SL DC Engine

Solution description



slinkin.tech



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What is SL DC Engine?

SL DC Engine is a Dissection and Classification Engine: a cross-platform and network independent DPI Engine. An integration solution written in modern C++.

Deep Packet Inspection (DPI) is a network traffic analysis technology that allows to extract their contents from packets, keep statistics on network flows and classify it, as well as filter and shape traffic.

DC Engine provides the ability to dissect packets, obtain protocol fields, collect statistics on network flows, and configure classification rules. DC Engine also makes it possible to expand the basic functionality through an extension mechanism that allows users to add their code to the traffic processing pipeline and use the results of its execution in classification rules.

The **DC Engine** is distributed as a modular software solution tailored to your specific network visibility, security, and performance requirements. Our commercial model consists of a permanent **Core** architecture combined with flexible, add-on **Extension Packages**. This decoupled approach ensures you only license the exact functionality your infrastructure demands, optimizing your Total Cost of Ownership (TCO) while allowing seamless scalability as your network evolves.

dc-engine core

- **Protocol detection.** Deep protocol analysis, field dissection, and decoding.
- **Metadata extraction.** Extracting packet/flow data (codecs, credentials, JA4 hashes).
- **Service classification.** Mapping traffic to internet services and types (media/files).
- **Flow statistics.** Tracking time, bytes, packets, bitrate, and IAT metrics.

> PACKAGES



service_update ready

Daily service classification configuration updates (data patterns, IPs, domains)

> protocols: **no**
> extensions: **probe**



security roadmap

ICMP/DNS tunnel detection, invalid certs, C&C server access, file extraction, leaks, DDoS

> protocols: **no**
> extensions: **sec, filecarver, sensdata, ddos**



industrial roadmap

The set of industrial protocols (detection, deep field dissection)

> protocols: **modbus, dnp3, profinet standard**
> extensions: **no**



workflow ready

Classification audio/video calls, chat, file transfer

> protocols: **no**
> extensions: **rtc_media, spid**



network roadmap

QoS estimation, error detection (re-transmissions, inactive flows, ping unreachables)

> protocols: **modbus, dnp3, profinet**
> extensions: **qos, netcon**



dlog roadmap

IPDR transmission to remote service/server (NetFlow, HTTP, JSON formats)

> protocols: **no**
> extensions: **dlog**

Functionality

SL DC Engine is a universal complex of network solutions, because its functional and technical features allow solving problems in the field of information security, system administration and telecommunications. The modular architecture of the product allows integration into third-party software solutions.

SL DC Engine has the following functionality:



Restricting access to prohibited resources



Prevention of exploitation of new vulnerabilities, before the release of official fixes (HotFix)



Maintaining logs of network activity that can be used in the investigation of incidents



Accumulation of information on streams for use in Billing



Load balancing in order to improve the quality of communication (traffic shaping)



Putting users under control (saving all or selected information for certain users)



Detection of network attacks (DDOS, Buffer Overflow, Scanning, etc.)



Integration into IT solutions (IPS/IDS, SIEM, DLP, etc.)



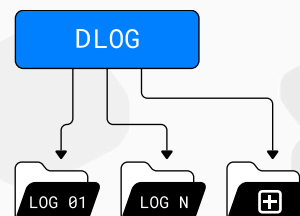
Data collection for AI systems

Technical advantages

> LOGGING SYSTEM

The logging module has a flexible configuration, which makes it possible to:

- Logging with the task of formatting and data for output
- Rotation of logs according to a schedule or file size
- Storing extracted field data to log journals for future analysis

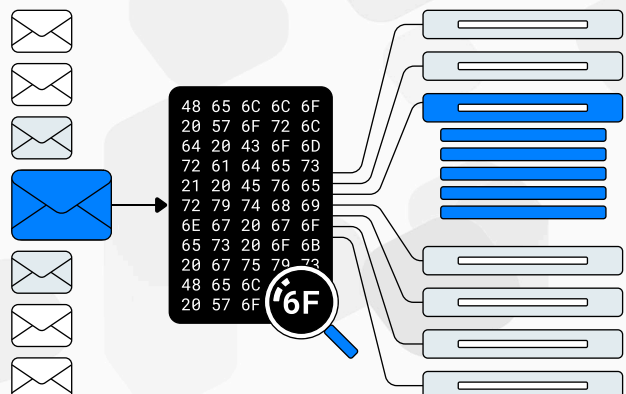


DLOG package only.

> DETAILED PACKET ANALYSIS

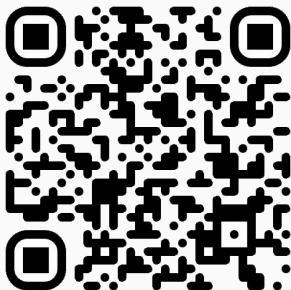
The packet analysis module is implemented with a number of advantages, for example, such as:

- The module does not allocate additional memory for packet fields (**in-place dissection**) and allows you to access them directly
- Cross-platform solution (C++)
- Convenient API
- Extension mechanism allows users/developers to extend engine functionality with own code based on other engine features
- Each supported protocol has its own documentation page
- Network independent (supported protocols of the Internet, IoT, Industrial stack)



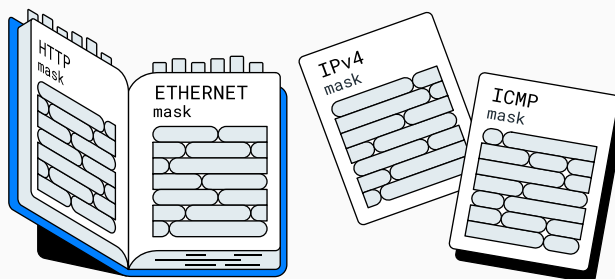
> PROTOCOLS

Our solution supports many network protocols, a list of which can be found on the page:



The protocol dictionary contains general information about each of them, and is also supported by the name of the RFC document according to which the protocol is supported. In addition, a table of supported fields is presented for each protocol, each of which has its own characteristics, such as identifier, name, type, size, mask, purpose.

When processing network traffic, the **SL DC Engine API** is used to dissect packets and gain access to protocol fields. This data can be used for logging and configuring restrictive policies. Third-party applications can use the result of the dissection to expand the functionality of their own applications, for example, using heuristic algorithms on the resulting data set or implementing complex logic to detect malicious activity on the network.



> INTEGRATION

The use of network traffic analysis technology is extremely popular for solutions such as **SEIM, IPS/IDS, DLP**. In addition to information security tasks, **DPI** is also used by Internet service providers and mobile operators to solve traffic classification problems, collect statistics, block prohibited resources and apply restrictions to subscribers.

SL DC Engine is an integration solution with a

user-friendly API. This allows developers to focus more on the logic and functionality of their product and spend less time implementing network traffic processing functions. **SL DC Engine** supports internal and external flow table mode.

> TRAFFIC FILTERING

Restricting access to prohibited resources and setting regulatory policies for network users are among the most common tasks in the field of network administration, network security analysis and prevention of network attacks. **SL DC Engine** is a solution that helps:

- Comply with the requirements of regulators to restrict access to Internet resources
- Register and analyze suspicious activity on the network in time
- Set limits for users on the speed and volume of transmitted information.

DPI technology helps to achieve maximum results when solving such tasks, providing the possibility of flexible configuration when setting filtering policies, as well as providing high performance when processing network traffic.



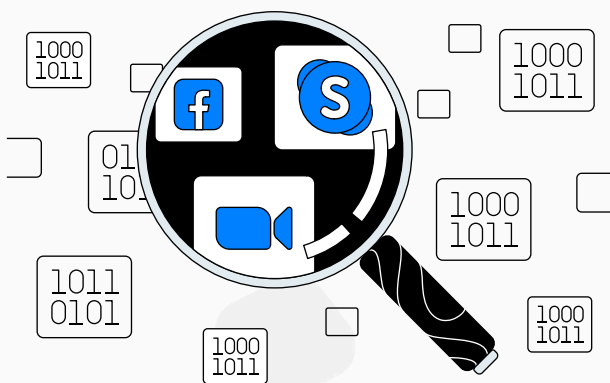
SL DC Engine does not provide blocking or shaping functionality; instead, it offers traffic classification for policy setup.

> CLASSIFICATION OF NETWORK SERVICES AND INTERNET SERVICES

The most important properties of network flows are:

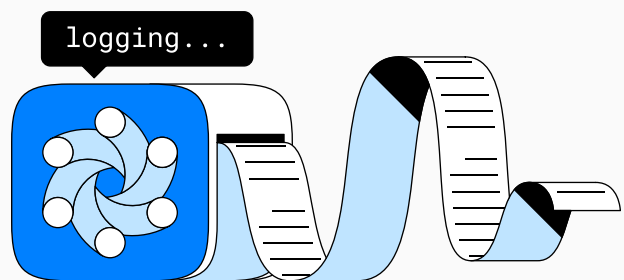
- Service definition
- Determining the nature of traffic (for example, audio/video traffic, file transfer, encryption, etc.)
- Determination of traffic properties (for example, gaming traffic, social networks, etc.)

DC Engine uses tagging mechanism for classification methods which helps to identify the service to which the network flow belongs. For example, Internet services, facebook, viber, youtube, etc. or network services such as ssh, telnet, dns, etc. Determining the properties of a flow is a complex technical task, since the underlying classification methods are mainly based on statistical indicators that may change in the process of updating/evolution of services. The tagging mechanism provides flexible functionality to mark network flows by any signature and behavior fingerprints which can be used not only for service determination but also for determining evil traffic, data leaks or just for marking flows for any other purposes (e.g. tethering, content type, etc.).



DC Engine provides a user-friendly API for working with classification capabilities, which helps:

- Update classification rules without changing the program code
- Add new services
- Easily transfer services and rules between releases/network nodes/applications
- All the above functionality is available via tagging mechanism configuration

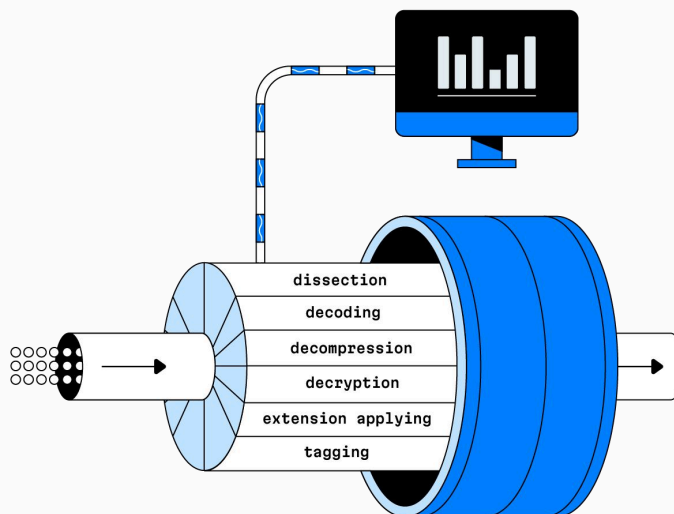


DC Engine keeps an internal event log available to the user, which helps to better understand the logic of the module, analyze the behavior, and understand on the basis of which characteristics the classification was made. This functionality significantly speeds up the process of mastering working with the module, and also helps to quickly identify problems, thereby reducing the time to fix them. The package includes detailed documentation on the supported protocols and their fields, built-in extensions (e.g. dns-cache) and **CLI** tool.

> FLEXIBILITY

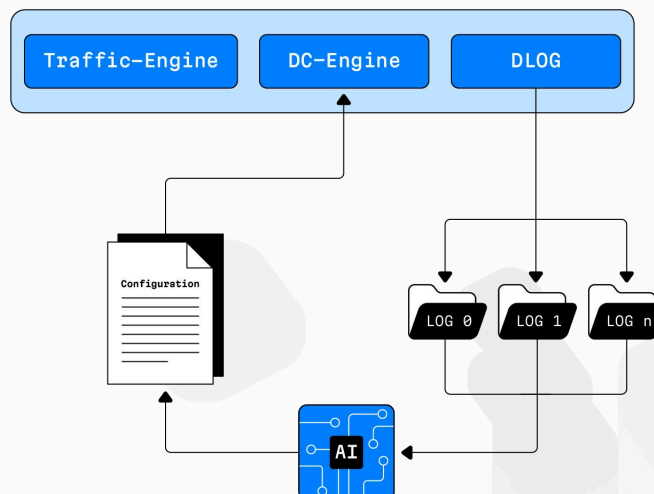
All solution packages are based on **DC Engine** and each of them is quite configurable. Configuration opportunities of engine and other solutions allow to set:

- Set thread count
- Set network interface settings
- Configure output journal format, log levels, log/journal rotation policies, log flush interval
- Configure packet assembling features such as fragment/segment count, max length, etc.
- Create tagging rules via human-readable language to classify network flows
- Create own code extensions to extend base engine functionality



> SELF-SUFFICIENT SYSTEM

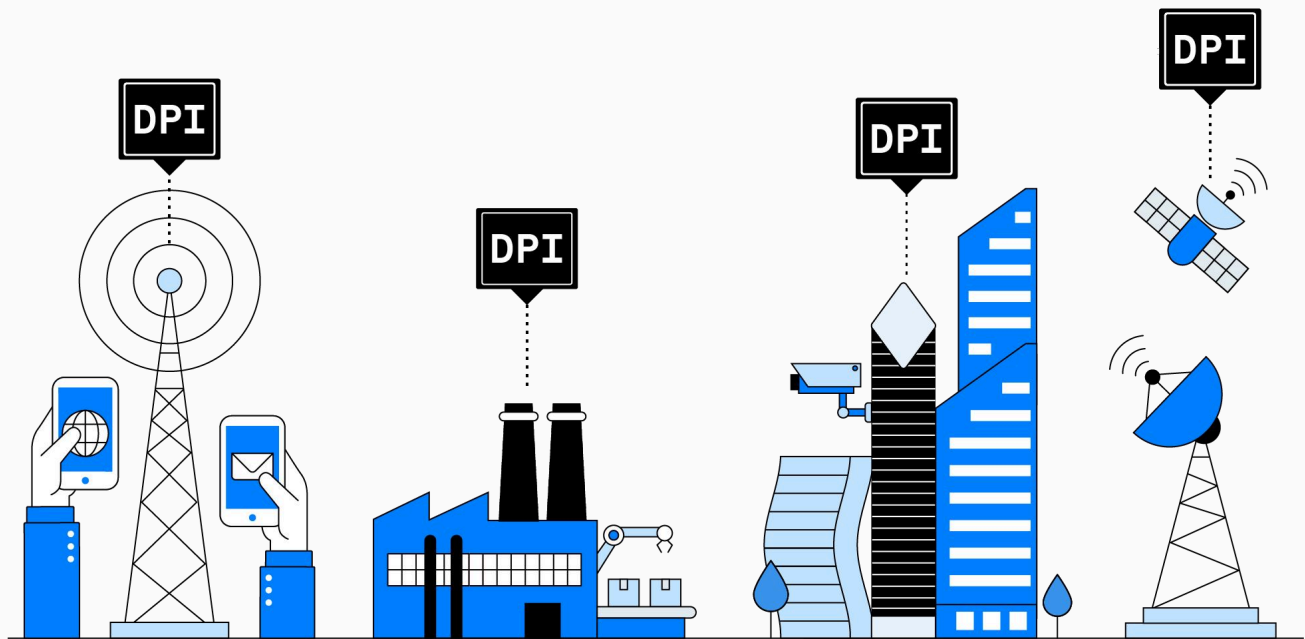
The **SL DC Engine** includes packages that work together to achieve optimal results in traffic classification. The **DLog** module logs data obtained from the **DC Engine**, including information extracted via user extensions. These log journals can then be analyzed by AI systems or cybersecurity analysts to detect new traffic patterns, which can be used to configure the **DC Engine** without any code changes.



> NETWORK INDEPENDENT

DPI plays a significant role in any kind of network – the Internet, IoT, Industrial, satellite, etc. That's because network monitoring is an important part of providing reliable and safe infrastructure. Network engineers and developers must have the opportunity to control the network for debugging and configuring purposes, and information

security engineers for detecting/preventing malicious activity. **DC Engine** is implemented as cross-platform and network-independent solution to be able to support protocols of any complexity level.



About us

Slinkin Technologies is an independent team of networking experts. We combine deep engineering expertise with an agile approach to deliver high-performance, flexible solutions for traffic analysis, balancing, and modification.

Even though our products function deep within network infrastructures, we take extra pride in making our complex tools as intuitive, reliable, and user-friendly as possible.

> CONTACTS



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